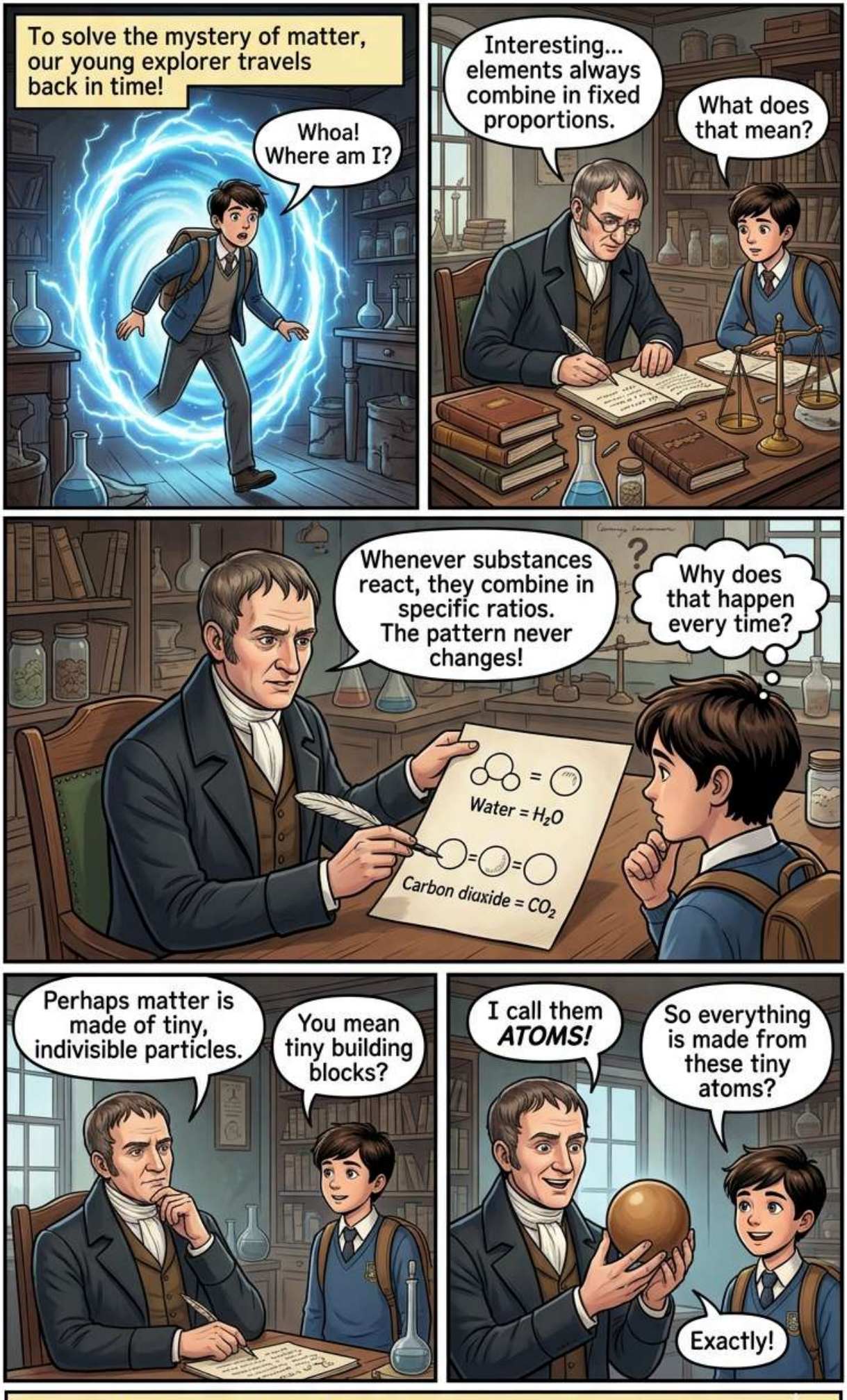
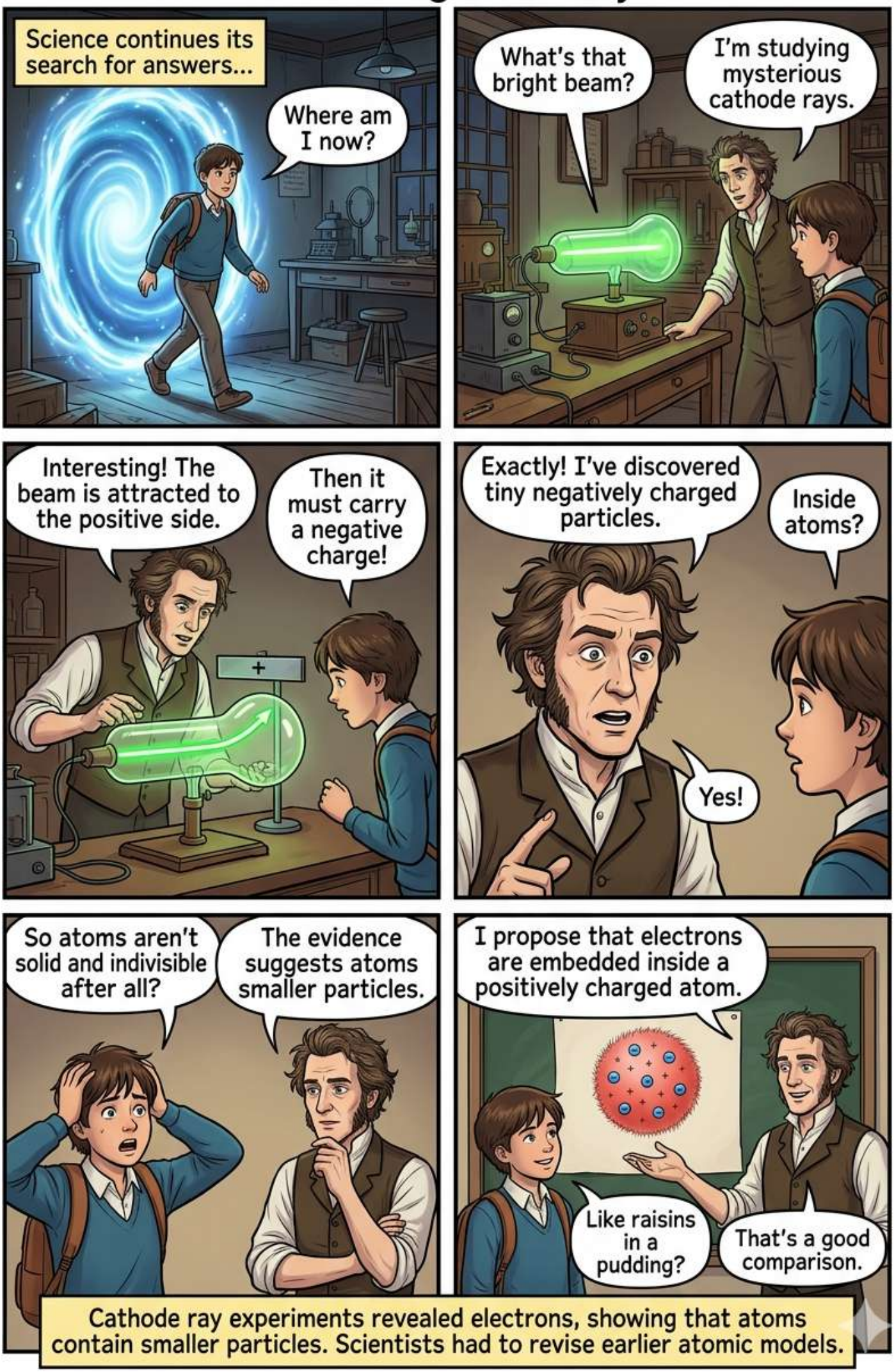

Everything in the universe is made of matter...
Everything looks different today...
What are these tiny things?
This rock...
...this water...
...even the air?
WHAT IS EVERYTHING MADE OF?
To find the answer, one student will travel through the greatest discoveries in science...
The Journey of Atomic Structure Begins.

The Journey of Atomic Structure



Reason for Dalton's Model: Scientists observed that elements always combined in fixed ratios. Dalton proposed atoms to explain this pattern.

A Shocking Discovery!

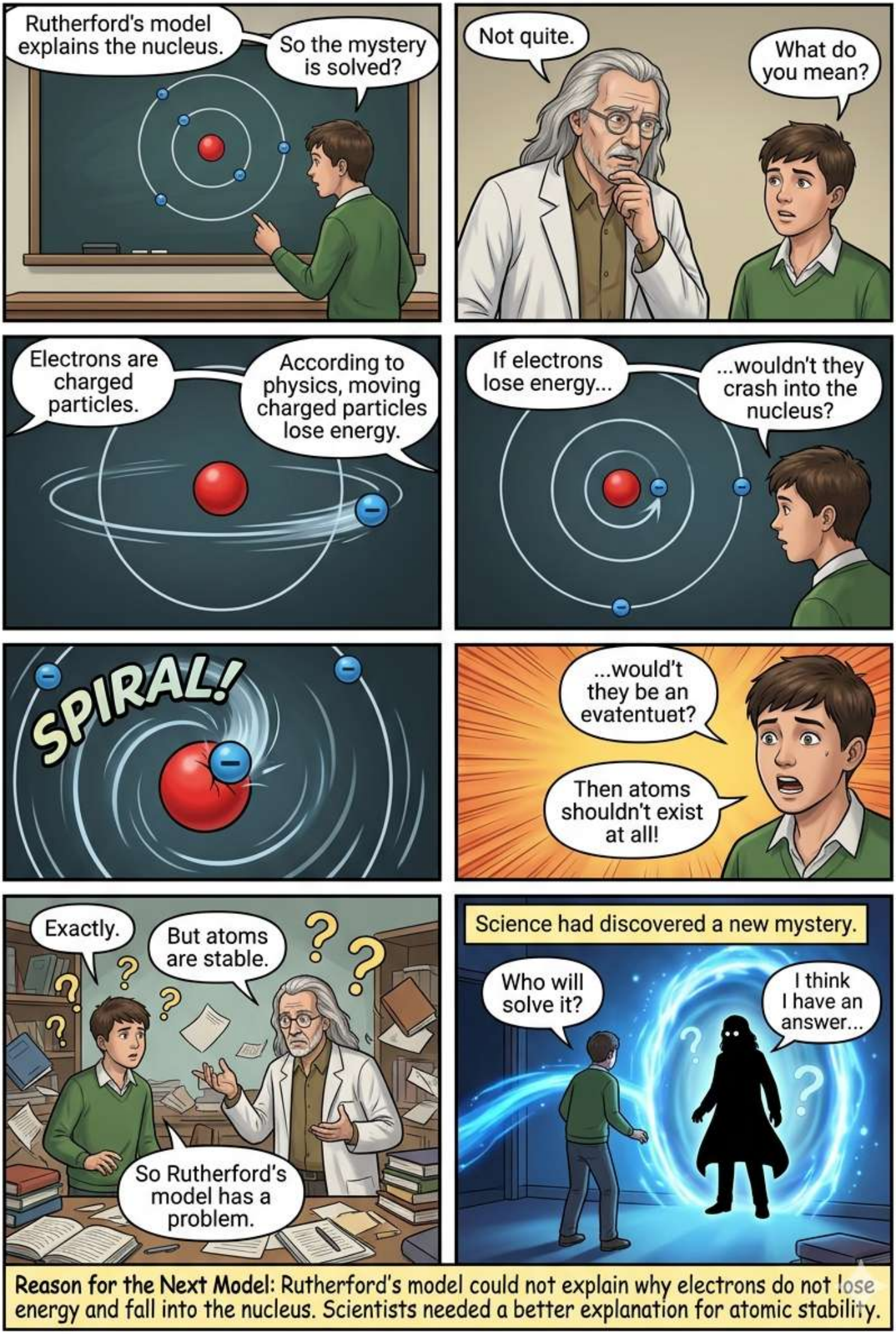


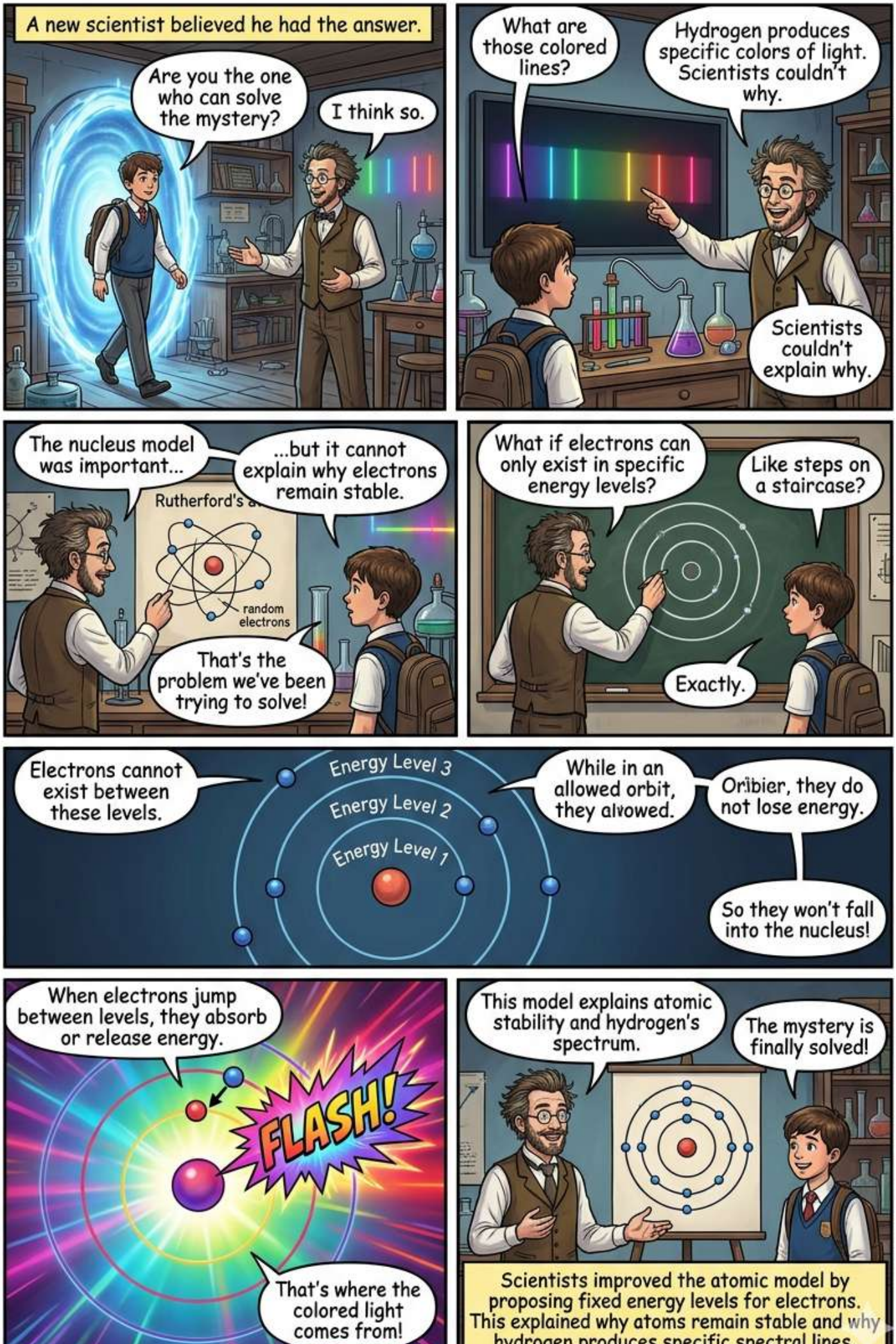
‘The Unexpected Bounce!’



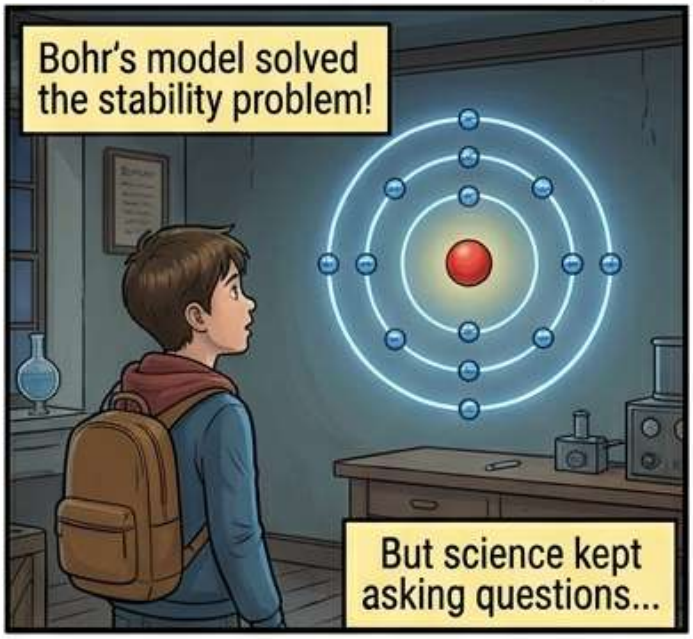
Reason for Rutherford's Model: Most alpha particles passed through the gold foil, but a few bounced back. This showed that atoms contain a tiny, dense, positively charged nucleus and are mostly empty space.

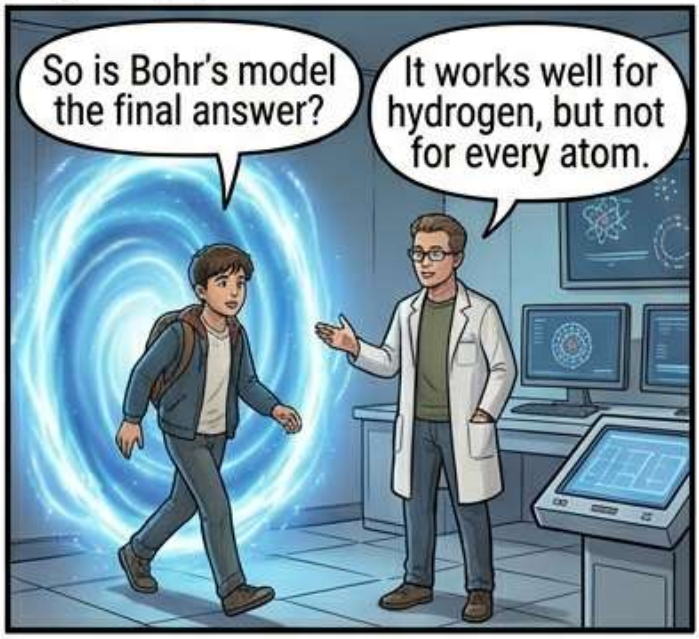
A Model in Trouble!

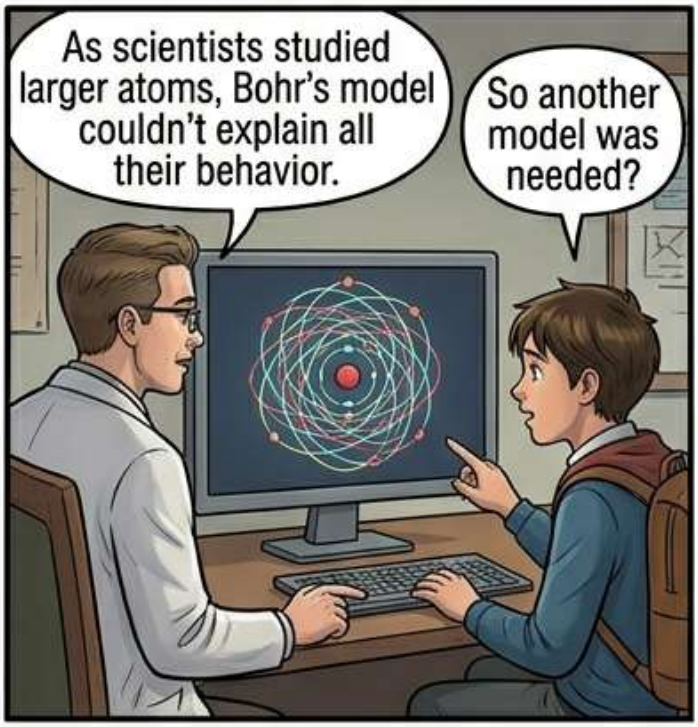


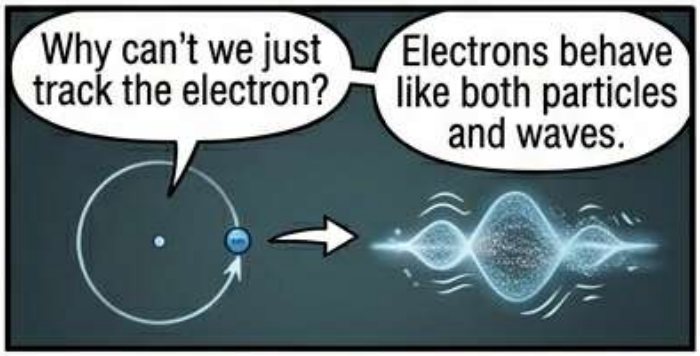

A new scientist believed he had the answer.
Are you the one who can solve the mystery?
I think so.
What are those colored lines?
Hydrogen produces specific colors of light. Scientists couldn't why.
Scientists couldn't explain why.
The nucleus model was important...
...but it cannot explain why electrons remain stable.
Rutherford's a
random electrons
That's the problem we've been trying to solve!
What if electrons can only exist in specific energy levels?
Like steps on a staircase?
Exactly.
Electrons cannot exist between these levels.
Energy Level 3
Energy Level 2
Energy Level 1
While in an allowed orbit, they alwowed.
Oribier, they do not lose energy.
So they won't fall into the nucleus!
When electrons jump between levels, they absorb or release energy.
FLASH!
That's where the colored light comes from!
This model explains atomic stability and hydrogen's spectrum.
The mystery is finally solved!
Scientists improved the atomic model by proposing fixed energy levels for electrons. This explained why atoms remain stable and why hydrogen produces specific spectral lines.

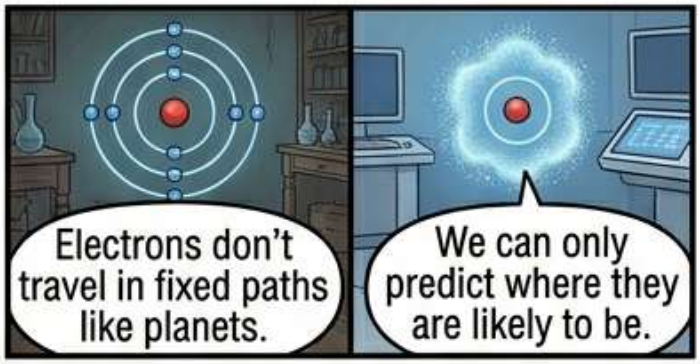
A New Mystery Appears!

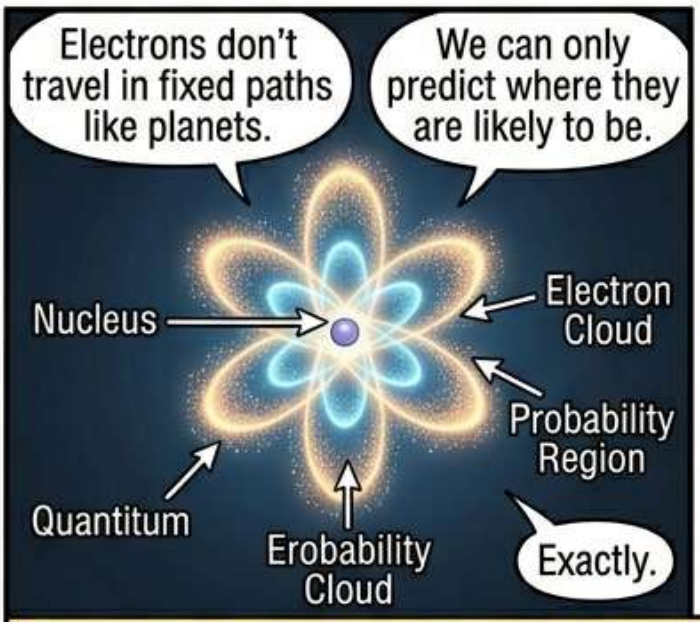


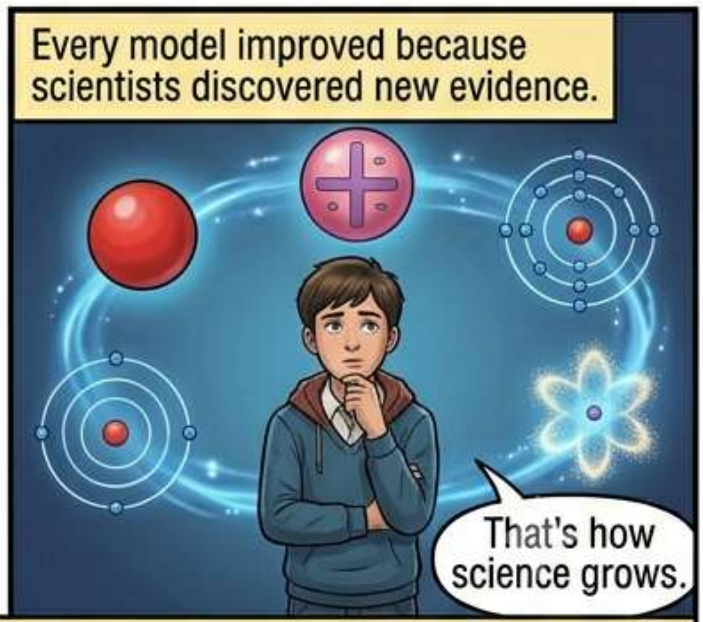




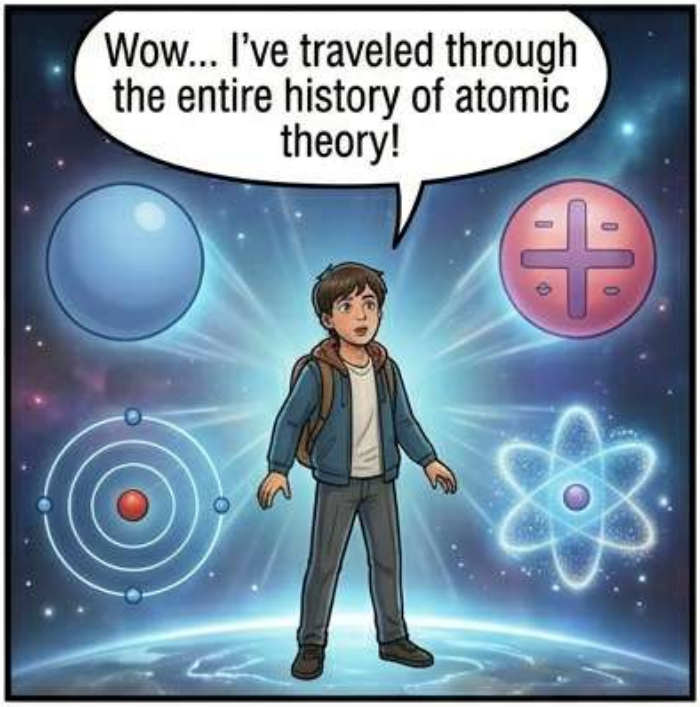


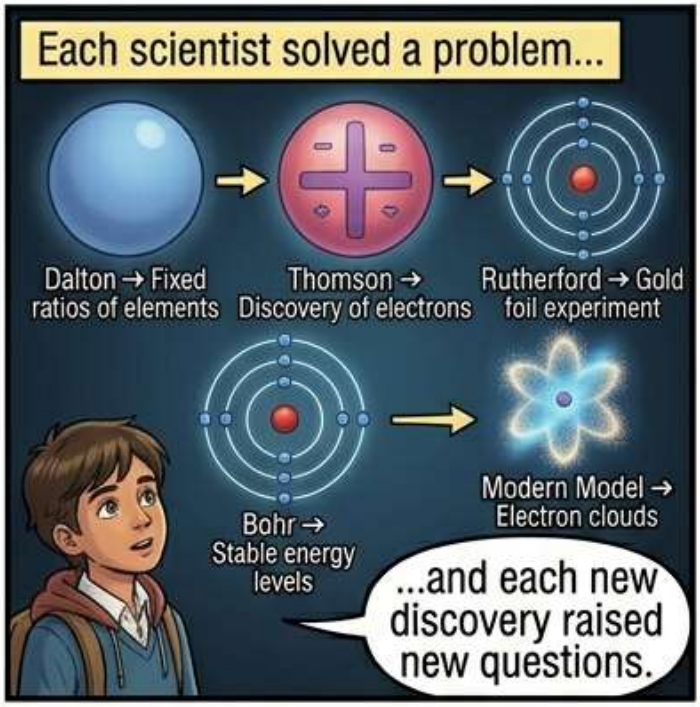






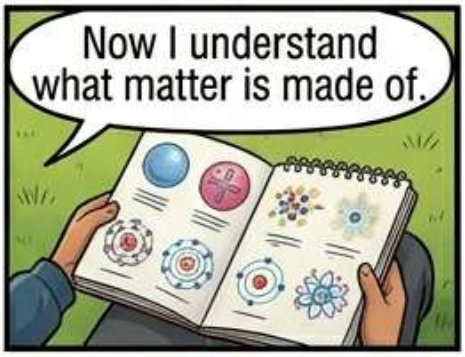
Reason for the Modern Quantum Model: Bohr's model could not fully explain complex atoms. Experiments showed that electrons behave like waves and particles, leading to the electron cloud model used today.

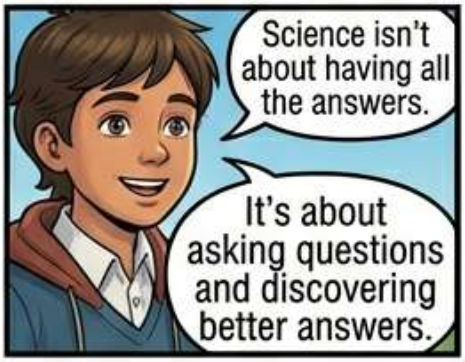

Wow... I've traveled through the entire history of atomic theory!

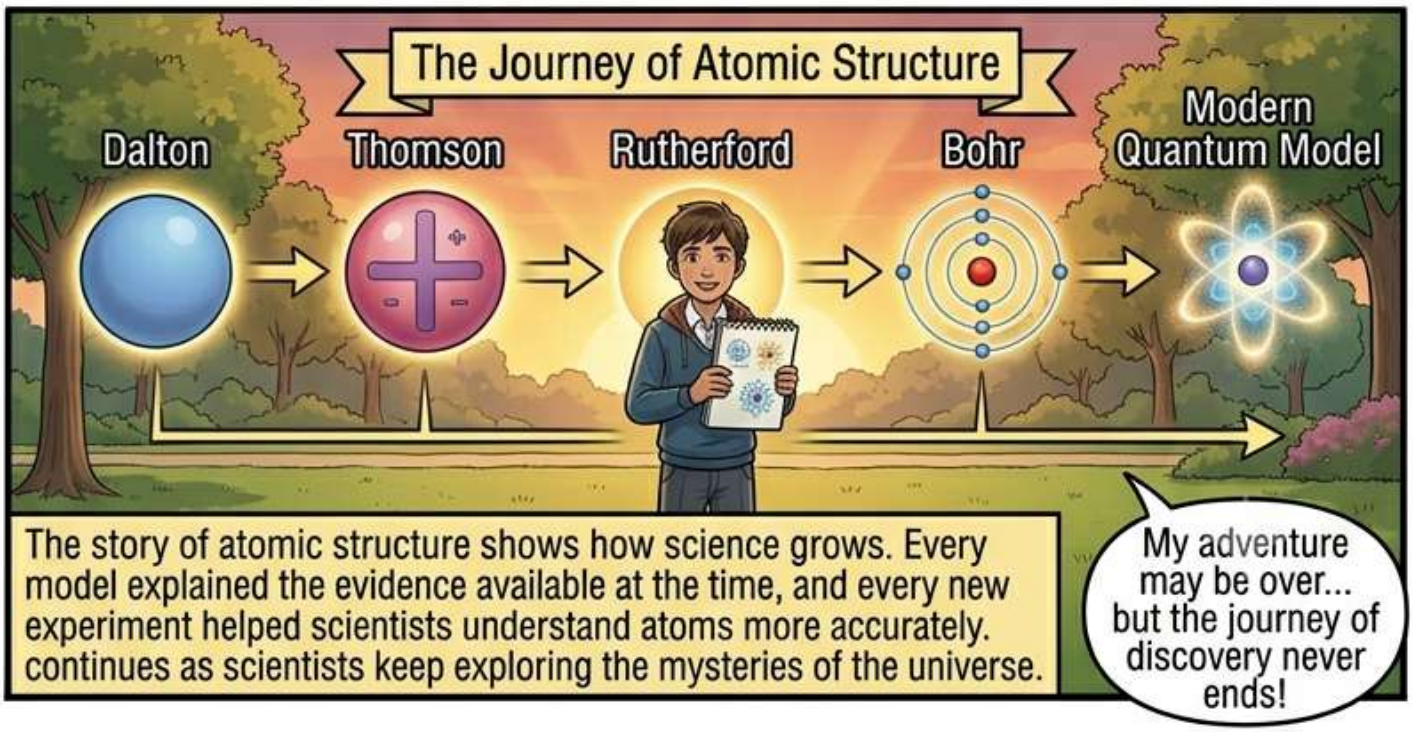

Each scientist solved a problem...
Dalton → Fixed ratios of elements
Thomson → Discovery of electrons
Rutherford → Gold foil experiment
Bohr → Stable energy levels
Modern Model → Electron clouds
...and each new discovery raised new questions.


The adventure was coming to an end.
I think it's time to go home.


I'm back where it all started!


Now I understand what matter is made of.


Science isn't about having all the answers.
It's about asking questions and discovering better answers.


The Journey of Atomic Structure
Dalton
Thomson
Rutherford
Bohr
Modern Quantum Model
The story of atomic structure shows how science grows. Every model explained the evidence available at the time, and every new experiment helped scientists understand atoms more accurately. continues as scientists keep exploring the mysteries of the universe.
My adventure may be over... but the journey of discovery never ends!